

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem
Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2856–2872; original printed pp. 95–96; physical source-PDF pp. 83–84; recomposed running pp. 75–76. The physical-page break occurs between French lines 2864 and 2865; the printed-page-96 marker occurs within line 2865 after *il existe*. Blank line 2873 is excluded; French line 2874 begins Section 3, “Applications,” and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration. The reference to diagram (2.7) is rendered literally, so this standalone file contains no unresolved reference. At line 2859 the corrected TeX underlines the prime ideal in prose but prints plain x in $\dim(A/x) = 1$; that source oddity is retained.

Suppose that $n > 1$, and assume the result established for every $n' < n$. By the induction hypothesis, for every $x \in U$ such that $c(x) = 1$, one has $x \notin \text{Ass } \mathcal{F}_x$. Let x be such a point, and let $y \in \overline{\{x\}} \cap Y$ be an immediate specialization of x , i.e.

$$\dim \mathcal{O}_{\overline{\{x\}}, y} = 1.$$

Perform the base change

$$v: \text{Spec}(\mathcal{O}_{X, y}) \longrightarrow X,$$

retaining the notation of diagram (2.7). Applying EGA III, 1.4.15, one obtains isomorphisms

$$v^*(R^p i_*(\mathcal{F}|_U)) \simeq R^p i'_*(v'^*(\mathcal{F}|_U)), \quad p \in \mathbb{Z}.$$

Thus we are reduced to the case in which X is the spectrum of a local ring A in which $\underline{x}$ is a prime ideal of dimension 1, i.e. $\dim(A/\underline{x}) = 1$. Set

$$\mathcal{F}' = \Gamma_Y(\mathcal{F}) \quad \text{and} \quad \mathcal{F}'' = \mathcal{F}/\mathcal{F}'.$$

We have $\mathcal{F}_x \simeq \mathcal{F}''_x$ and $y \notin \text{Ass } \mathcal{F}''$. Moreover, $\mathcal{F}'|_U = 0$; hence the long exact sequence of the $R^p i_*$ gives isomorphisms

$$R^p i_*(\mathcal{F}|_U) \simeq R^p i_*(\mathcal{F}''|_U), \quad p \in \mathbb{Z}.$$

French printed p. 96 Since $n > 1$, neither x nor y belongs to $\text{Ass } \mathcal{F}''$. Now $\underline{x}$ and y are the only prime ideals of A containing $\underline{x}$. It follows from Exposé III, 2.1 that there exists an element $g \in \underline{x}$ that is M -regular, where we have set

$$\mathcal{F} = \widetilde{M}, \quad M = \mathcal{F}(X).$$

Thus there is an exact sequence

$$0 \longrightarrow M \xrightarrow{g'} M \longrightarrow N \longrightarrow 0,$$

in which g' denotes multiplication by g on M . From the resulting long exact sequence one sees that

$$R^p i_*(\widetilde{N}|_U) \text{ is coherent for } p < n - 1.$$

Therefore, by the induction hypothesis,

$$\text{prof}(\widetilde{N})_x \geq n - 1, \quad \text{and hence} \quad \text{prof } \mathcal{F}_x \geq n.$$

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